JGH Open
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An open access journal of gastroenterology and hepatology
JGHF
Check for updates
doi:10.1002/jgh3.12877

ORIGINAL ARTICLE

Prevalence, association, and prognostic significance of polypharmacy and sarcopenia in patients with liver cirrhosis

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Key words
cirrhosis, multimorbidity, older adults, polypharmacy, sarcopenia.

Accepted for publication 1 February 2023.

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Author contribution: All authors contributed to the conception and design of this study. Data collection and analysis were performed by Tatsunori Hanai, Kayoko Nishimura, Takao Miwa, Toshihide Maeda, Kenji Imai, Atsushi Suetsugu, and Koji Takai. The first draft of the manuscript was written by Tatsunori Hanai, and all authors commented on the previous versions of the manuscript. All authors have read and approved the final manuscript.
Declaration of conflict of interest: The authors declare no conflict of interest.
Financial support: This work was supported by a Grant-in-Aid for Research from the Japan Agency for Medical Research and Development (JP22fk0210113).

Funding support: Grant-in-Aid for Research from the Japan Agency for Medical Research and DevelopmentJP22fk0210113

Abstract

Background and Aim: Polypharmacy and sarcopenia are increasing public health problems worldwide. However, data on the prevalence, association, and prognostic significance of polypharmacy and sarcopenia in patients with liver cirrhosis are limited.
Methods: Polypharmacy and sarcopenia were assessed in 239 patients with liver cirrhosis. Polypharmacy was defined as the daily use of six or more medications, and sarcopenia was diagnosed based on muscle strength and mass evaluated on computed tomography. The association between polypharmacy and sarcopenia and their effects on mortality were analyzed using logistic regression and Cox proportional hazards models.
Results: Among the 239 patients, 52% were men, the median age was 68 years, and the number of medications used per patient was 6. Further, 53% and 29% patients had polypharmacy and sarcopenia, respectively. The number of medications used and the prevalence of sarcopenia increased with age. Patients with polypharmacy and sarcopenia had similar characteristics, such as older age, increased medication use, advanced liver disease, and decreased muscle strength and mass. After adjusting for confounders, polypharmacy was significantly associated with sarcopenia (odds ratio, 2.11; 95% confidence interval [CI], 1.07–4.17). During the median follow-up of 2.2 years, 62 (26%) patients died. Polypharmacy (hazard ratio [HR], 1.83; 95% CI, 1.01–3.37) and sarcopenia (HR, 2.00; 95% CI, 1.12–3.50) independently predicted mortality. The prognostic significance of polypharmacy was more prominent in older adults than in younger adults (HR, 2.31; 95% CI, 1.01–5.67).
Conclusion: Polypharmacy and sarcopenia are interrelated and associated with poor prognosis in patients with cirrhosis. Further large, prospective, population-based studies are required to validate these findings.

Introduction

Polypharmacy and sarcopenia are serious public health problems worldwide, given the rapid increase in the older adult population.[1–3] Owing to the dramatic increase in life expectancy in the 20th century, the number of people aged ≥65 years is expected to increase from an estimated 524 million (8% of the world population) in 2010 to approximately 1.5 billion (16%) in 2050.[2,3] Japan is one of countries with the largest aged population in the world, with an average life expectancy of >83 years at birth.[3,4] In addition, 27% of the Japanese population was aged ≥65 years in 2016, and this is expected to reach 40% by 2060.[4,5] Reflecting the aging population, the number of older adults with liver cirrhosis has been increasing in Japan, with the average age of these patients increasing from 63.8 years in 2007 to 68.1 years in 2014.[6,7]

Polypharmacy—the concurrent use of multiple medications—is common among older adults owing to the increased risk of age-related complications, including chronic diseases, multimorbidity, and geriatric syndromes.[1,2,8] Aging-related decreases in renal clearance, liver function, and skeletal muscle mass affect drug pharmacokinetics and pharmacodynamics.[1,8,9] Polypharmacy has

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doi:10.1002/jgh3.12877

原始文章

肝硬化患者多药和肌萎缩的患病率、关联及预后意义

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关键词
肝硬化、多种合并症、老年患者、多重用药、肌肉减少症

2023年2月1日接受发表。

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作者贡献：本研究的构思与设计由全体作者共同完成。数据收集与分析工作由花井达纪、西村香代子、三和隆夫、前田俊秀、今井贤治、铃木敦司及高井浩司共同完成。初稿由花井达纪执笔，所有作者均参与了稿件修订。最终定稿经全体作者审阅并签署同意。

利益冲突声明：作者声明不存在利益冲突。

资金支持：本研究获得日本医疗研究开发机构（JP 22fk0210113）研究资助。

资助信息：日本医疗研究开发机构研究资助项目 JP22fk0210113

摘要

背景与目的：多药并用和肌少症是全球范围内日益严重的公共卫生问题。然而，关于肝硬化患者中这两种病症的患病率、关联性及其预后意义的数据仍十分有限。

研究方法：本研究对239例肝硬化患者进行多药联用与肌肉减少症评估。多药联用定义为每日使用六种及以上药物，肌肉减少症通过计算机断层扫描评估肌肉力量和质量确诊。采用逻辑回归模型和Cox比例风险模型，分析多药联用与肌肉减少症对死亡率的影响。

研究结果显示：在239名患者中，男性占52%，中位年龄为68岁，每位患者平均使用6种药物。此外，53%和29%的患者分别存在多药联用和肌少症。药物使用量与肌少症患病率均随年龄增长而增加。多药联用和肌少症患者具有相似特征，包括年龄较大、用药量增加、肝病进展严重以及肌肉力量和质量下降。经混杂因素校正后，多药联用与肌少症显著相关（比值比2.11；95%置信区间[CI] 1.07-4.17）。在中位随访期间
2.2年随访期间，62例（26%）患者死亡。多药并用（风险比[HR] 1.83；95%置信区间 1.01-3.37）和肌少症（HR 2.00；95%置信区间 1.12-3.50）是独立的死亡预测因素。多药并用对老年患者的预后影响显著高于年轻患者（HR 2.31；95%置信区间 1.01-5.67）。

结论：多药并用与肌少症密切相关，二者均与肝硬化患者预后不良相关。需开展更多大规模前瞻性人群研究以验证这些发现。

介绍

随着全球老年人口的快速增长，多重用药和肌肉减少症已成为严重的公共卫生问题。[1–3] 由于20世纪人类预期寿命的显著提升，65岁及以上人口数量预计将从2010年的约5.24亿（占全球人口的8%）激增至2050年的15亿（占16%）。[2,3] 日本作为全球老龄化程度最高的国家之一，其新生儿平均预期寿命已超过83岁。[3,4] 此外，日本65岁以上人口占比高达27%。

2016年，这一比例预计到2060年将达到40%。[4,5] 反映人口老龄化，日本肝硬化老年人数持续增加，这些患者的平均年龄从2007年的63.8岁上升到2014年的68.1岁。[6,7]

老年患者常出现多药并用现象，这主要与年龄相关并发症风险升高有关，包括慢性疾病、多重病症及老年综合征。[1,2,8] 随着年龄增长，肾清除率、肝功能及骨骼肌质量下降，这些变化会影响药物的药代动力学和药效学。[1,8,9] 多药并用现象

recently attracted attention because it is associated with adverse drug reactions, drug–drug and drug–disease interactions, poor treatment adherence, decreased drug efficacy, falls, cognitive impairment, sarcopenia, impaired activities of daily living, increased hospitalization, and mortality.[1,2,8] In addition, patients with cirrhosis are at an increased risk of polypharmacy because of complications associated with cirrhosis.[10]

Sarcopenia, a progressive and generalized skeletal muscle disorder characterized by loss of skeletal muscle mass and strength, is a well-known geriatric syndrome.[1] Sarcopenia is common in not only older adults but also in patients with cancers and chronic diseases and is associated with adverse clinical outcomes, including falls, fractures, hospitalization, impaired activities of daily living, cognitive impairment, reduced quality of life, and mortality.[11,12] Substantial evidence shows that the prevalence of sarcopenia is higher in patients with cirrhosis than in the older adult population and in patients with other diseases such as heart failure, renal failure, chronic obstructive pulmonary disease, and cancer.[11] Therefore, patients with cirrhosis may be at increased risk of polypharmacy and sarcopenia due to age-related physiological changes, multimorbidity, and impaired liver function.[6,10,13]

Therefore, there is an urgent need to identify patients with cirrhosis at the risk of polypharmacy and sarcopenia, determine the prognostic significance of these factors, and provide appropriate pharmacotherapy to improve clinical outcomes. However, few studies have evaluated the prevalence, association, and prognostic significance of polypharmacy and sarcopenia, especially in older adults with cirrhosis. This study aimed to evaluate the characteristics of older adults with liver cirrhosis, prevalence and association of polypharmacy and sarcopenia, and the extent to which these factors affect the prognosis of these patients.

Patients and methods

Study design and participants. In total, 239 patients with cirrhosis treated at Gifu University Hospital (Gifu, Japan) between October 2011 and December 2020 were enrolled in this retrospective cohort study. The observation period was from the date of enrollment to the date of last visit, death, or December 31, 2020, whichever occurred first. The study protocol was reviewed and approved by the Institutional Review Board of Gifu University Graduate School of Medicine (approval number: 2021-B190). This study was performed in accordance with the ethical standards of the Declaration of Helsinki 1964 and its later amendments.

Data collection. Patients were categorized into older adult (age ≥65 years) and younger adult groups (age <65 years).[3–5] Data on the number of medications used and demographic and clinical characteristics of patients within 1 month of sarcopenia assessment were collected from the prospective database using a standard template. Data on the following demographic and clinical characteristics were collected: age, sex, number of medications used, polypharmacy, height, weight, body mass index (BMI), grip strength, skeletal muscle mass, sarcopenia, etiology of cirrhosis, ascites, hepatic encephalopathy, Child–Pugh score, Model for End-stage Liver Disease (MELD) score, and laboratory findings. Liver cirrhosis was diagnosed based on laboratory findings, clinical features (spider nevus, caput medusae, esophagogastric varices, and portosystemic shunt), and imaging and histological features. The inclusion criteria were age ≥20 years, cirrhosis of any etiology, and assessment of sarcopenia and polypharmacy. The exclusion criteria were a history of organ transplantation, hepatocellular carcinoma, active non-hepatic malignancies, amino acid metabolism disorder, pregnancy, neurological or orthopedic diseases that affect grip strength measurement, and medical conditions such as severe sepsis, heart failure, respiratory failure, renal failure, and other acute life-threatening diseases.

Polypharmacy. Data on medication use were recorded in electronic records by pharmacists who were not directly involved in patient care. Polypharmacy was assessed based on the number of oral medications used for at least 28 days within 1 month of sarcopenia assessment. Polypharmacy was defined as the daily use of six or more medications based on the criteria established by the Ministry of Health, Labour and Welfare. Polypharmacy is associated with an increased risk of adverse drug reactions in the older adult population in Japan.[14,15]

Sarcopenia. Sarcopenia was diagnosed based on the revised sarcopenia criteria for liver disease (second edition) proposed by the Japanese Society of Hepatology,[16] including grip strength and skeletal muscle mass. Patients who met both criteria were diagnosed with sarcopenia. Grip strength was measured by a registered dietitian using a digital Smedley dynamometer (T.K.K.5101 GRIP-D; Takei Scientific Instruments, Niigata, Japan), and the cutoff value for low grip strength was 28 kg for men and 18 kg for women. Skeletal muscle area was measured by estimating the cross-sectional area of the abdominal skeletal muscles at the third lumbar vertebra using a computed tomography image analysis system (Synapse Vincent; Fujifilm, Tokyo, Japan). Skeletal muscle index (SMI) was calculated by dividing the skeletal muscle area by the square of the height, and the cutoff value for low SMI was 42 cm^2/m^2 for men and 38 cm^2/m^2 for women.

Statistical analysis. Continuous variables are presented as the median and interquartile range (IQR), and groups were compared using the Mann–Whitney *U*-test or the Kruskal–Wallis test. The normality of the distribution of continuous variables was evaluated using the Shapiro–Wilk normality test. Categorical variables are presented as the number of patients and percentage, and groups were compared using the chi-square test or Fisher's exact test. The association between polypharmacy and sarcopenia was assessed using the logistic regression model, and the results are presented as odds ratios (ORs) and 95% confidence intervals (CIs). The prognostic significance of polypharmacy and sarcopenia was assessed using the Cox proportional hazards model, and the results are presented as hazard ratios (HRs) and 95% CIs. Overall survival (OS) was estimated using the Kaplan–Meier method, and the groups were compared using the log-rank test. All tests were two sided, and the significance threshold was set at $P < 0.05$. All analyses were performed using JMP software, version 9.0.2 (SAS Institute Inc., Cary, NC, USA).

Results

Patient characteristics. Among the 239 patients included, 125 (52%) were men, with a median age of 68 years, BMI of

近年来，肝硬化因其与药物不良反应、药物间及药物与疾病间的相互作用、治疗依从性差、药物疗效降低、跌倒、认知障碍、肌肉减少症、日常生活活动能力受损、住院率升高及死亡率增加等关联而备受关注。[1,2,8] 此外，由于肝硬化相关并发症，患者还面临多重用药风险。[10]

肌少症是一种以骨骼肌质量与力量逐渐丧失为特征的进行性全身性骨骼肌疾病，是老年群体中常见的健康问题。[1] 这种病症不仅普遍存在于老年人群，癌症患者和慢性病患者中也较为常见，与跌倒、骨折、住院治疗、日常生活能力下降、认知功能障碍、生活质量降低及死亡率上升等不良临床结局密切相关。大量研究数据表明，肝硬化患者的肌少症患病率显著高于普通老年人群，且在心力衰竭、肾功能衰竭、慢性阻塞性肺病和癌症等其他疾病患者中更为突出。[11] 因此，肝硬化患者由于年龄相关生理变化、多重疾病共存及肝功能受损等因素，可能面临更高的多重用药风险和肌少症发生概率。[11,12 6,10,13]

因此，亟需识别存在多药联用和肌少症风险的肝硬化患者，明确这些因素的预后意义，并提供适宜的药物治疗以改善临床结局。然而，目前鲜有研究评估多药联用与肌少症的患病率、关联性及预后意义，特别是在老年肝硬化患者中。本研究旨在评估老年肝硬化患者的特征、多药联用与肌少症的患病率及关联性，以及这些因素对患者预后的影响程度。

患者与方法

研究设计与受试者。

共计239例患者本回顾性队列研究纳入2011年10月至2020年12月期间在日本岐阜大学医院（岐阜，日本）接受治疗的肝硬化患者。观察期自入组之日起至末次访视、死亡或2020年12月31日（以先发生者为准）结束。研究方案经岐阜大学医学研究生院机构审查委员会审核批准（批准号：2021-B190）。本研究严格遵循1964年《赫尔辛基宣言》及其后续修正案的伦理标准。

数据收集。

患者被归类为老年成人研究对象分为老年组（≥65岁）和年轻成人组（<65岁）。研究人员通过标准模板从前瞻性数据库中提取了患者在肌少症评估前1个月内的用药数据及人口统计学与临床特征信息，具体包括：年龄、性别、用药数量、多重用药情况、身高、体重、体重指数（BMI）、握力、骨骼肌质量、肌少症状态、肝硬化病因、腹水情况、肝性脑病表现、Child-Pugh评分、终末期肝病模型（MELD）评分以及实验室检查结果。肝硬化诊断依据实验室检测结果和临床特征（如蜘蛛痣、头端静脉曲张、食管胃静脉曲张等），（包括门体分流）以及影像学和组织学特征。纳入标准为年龄≥20岁、任何病因的肝硬化，以及肌少症和多重用药的评估。排除标准包括器官移植史、肝细胞癌、活动性非肝恶性肿瘤、氨基酸代谢障碍、妊娠、影响握力测量的神经或骨科疾病，以及严重脓毒症、心力衰竭、呼吸衰竭、肾衰竭和其他急性危及生命疾病等医学状况。

多重用药。

用药数据记录于由未直接参与患者护理的药师管理电子病历。多药共用情况根据肌少症评估后1个月内使用至少28天的口服药物数量进行评估。根据日本厚生劳动省制定的标准，多药共用定义为每日使用六种或以上药物。在日本老年人群中，多药共用与药物不良反应风险增加相关。[14,15]

肌肉减少症。

肌肉减少症的诊断依据为修订版日本肝病学会制定的肝病患者肌肉减少症诊断标准（第二版）[16] 包括握力和骨骼肌质量两项指标。符合两项标准的患者即可确诊肌肉减少症。握力检测由注册营养师使用数字式Smedley测力计（型号T.K.K.5101GRIP-D，日本新潟竹井科学仪器公司）完成，男性最低标准为28公斤，女性为18公斤。骨骼肌面积通过CT图像分析系统（SynapseVincent，日本富士胶片公司）测量腰椎第三节横截面的腹部骨骼肌面积确定。骨骼肌指数（SMI）计算公式为骨骼肌面积除以身高平方，男性 ^{2} 的低值标准为42 ^{2} 厘米/米，女性则为38厘米 ^{2} /米 ^{2} 。

统计分析。

连续变量以中位数和四分位距（IQR），组间比较采用曼-惠特尼U检验或克鲁斯卡尔-沃利斯检验。连续变量的正态性通过夏皮罗-威尔克正态性检验评估。分类变量以患者数量和百分比表示，组间比较采用卡方检验或费希尔精确检验。多药治疗与肌少症之间的关联通过逻辑回归模型评估，结果以比值比（ORs）和95%置信区间（CIs）呈现。多药治疗和肌少症的预后意义通过Cox比例风险模型评估，结果以风险比（HRs）和95%置信区间（CIs）呈现。总生存期（OS）采用Kaplan-Meier法估计，组间比较采用对数秩检验。所有检验均为双侧检验，显著性阈值设定为 $P<0.05$。所有分析均使用JMP 软件（版本9.0.2，SAS研究所，美国北卡罗来纳州卡里）完成。

结果

患者特征。

在纳入的239例患者中，125 （52%）为男性，中位年龄68岁，BMI为

22.7 kg/m^2, Child–Pugh score of 7, and MELD score of 9. Altogether, 127 (53%) and 69 (29%) patients had polypharmacy and sarcopenia, respectively. The median number of medications used by each patient was 6 (IQR, 4–8, Table 1). The number of medications increased significantly with age ($P < 0.037$; Fig. 1a). Similarly, the prevalence of polypharmacy tended to increase with age ($P = 0.153$; Fig. 1b). The types of medication used are presented in Table S1. Seven patients who received beta-blockers to prevent variceal bleeding were classified in the antihypertensive group.

In total, 148 (62%) and 91 (28%) patients were categorized into the older and younger adult group, respectively (Table 1). Compared to the younger adult group, the older adult group had more patients with polypharmacy, high serum creatinine levels, and viral cirrhosis, but fewer patients with alcoholic cirrhosis. Height and weight were significantly lower in the older adult group than in the younger adult group; however, there was no significant intergroup difference in BMI. Grip strength and SMI were significantly lower in the older adult group than in the younger adult group, and sarcopenia was more predominant in the older adult group than in the younger adult group (37% vs 15%, $P < 0.001$; Table 1). The prevalence of sarcopenia increased with age ($P < 0.001$; Fig. 1c).

Polypharmacy and sarcopenia in patients with cirrhosis. The prevalence of sarcopenia was significantly higher in patients with polypharmacy than those without polypharmacy (36% vs 21%; $P = 0.010$). Similarly, the prevalence of polypharmacy was higher in patients with sarcopenia than those without sarcopenia (67% vs 48%; $P = 0.010$). The patients with polypharmacy and those with sarcopenia had similar characteristics, such as older age, increased medication use, advanced liver disease, and lower grip strength and SMI (Table 2).

Association between polypharmacy and sarcopenia. The unadjusted OR for polypharmacy in sarcopenia was 2.20 (95% CI, 1.23–3.94; $P = 0.008$) (Table 3). After adjusting for age, sex, BMI, etiology of cirrhosis, and MELD score, polypharmacy remained significantly associated with sarcopenia (OR, 2.11; 95% CI, 1.07–4.17; $P = 0.032$). This association was more pronounced in the older adult group (OR, 2.62;

Table 1 Baseline characteristics of patients with cirrhosis

Characteristic	Total cohort ($n = 239$)	Younger adults ($n = 91$)	Older adults ($n = 148$)	*P*-value*
Age, years	68 (58–76)	55 (50–60)	74 (70–79)	<0.001
Men	125 (52)	53 (58)	72 (49)	0.182
Number of medications used	6 (4–8)	5 (4–7)	7 (3–9)	0.069
Polypharmacy	127 (53)	41 (45)	86 (58)	0.049
Comorbidities				
Diabetes mellitus	66 (28)	23 (25)	43 (29)	0.555
Hypertension	151 (63)	51 (56)	100 (68)	0.097
Dyslipidemia	6 (3)	2 (2)	4 (3)	0.809
Etiology of cirrhosis				
Hepatitis B virus	17 (7)	3 (3)	14 (10)	0.002
Hepatitis C virus	68 (29)	19 (21)	49 (33)	
Alcoholic	60 (25)	34 (37)	26 (18)	
Others	94 (39)	35 (39)	59 (40)	
Ascites	113 (47)	44 (48)	69 (47)	0.894
Hepatic encephalopathy	23 (10)	7 (8)	16 (11)	0.503
MELD score	9 (7–12)	10 (8–12)	9 (7–12)	0.155
Child–Pugh score	7 (5–9)	7 (6–9)	7 (5–9)	0.525
Albumin, g/dl	3.2 (2.6–3.7)	3.2 (2.5–3.5)	3.2 (2.6–3.8)	0.454
Creatinine, mg/dl	0.71 (0.60–0.90)	0.65 (0.52–0.78)	0.74 (0.63–0.93)	0.002
Sodium, mEq/L	138 (136–140)	138 (137–140)	138 (136–140)	0.401
Total bilirubin, mg/dl	1.2 (0.8–1.7)	1.3 (0.9–1.9)	1.1 (0.8–1.6)	0.057
International normalized ratio	1.12 (1.03–1.26)	1.18 (1.04–1.31)	1.10 (1.02–1.23)	0.008
Body composition				
Height, cm	158 (150–166)	164 (158–171)	155 (149–161)	<0.001
Weight, kg	58.3 (50.3–65.9)	63.7 (53.9–74.6)	56.0 (48.8–62.5)	<0.001
Body mass index, kg/m^2	22.7 (21.4–25.4)	22.73 (21.8–26.3)	22.7 (21.0–25.1)	0.238
Grip strength, kg	20.9 (16.5–29.2)	26.8 (19.7–35.5)	18.6 (14.5–24.9)	<0.001
Skeletal muscle index, cm^2/m^2	41.9 (36.2–49.1)	44.9 (38.1–50.0)	40.1 (35.7–47.8)	0.011
Sarcopenia	69 (29)	14 (15)	55 (37)	<0.001

Note: Values are presented as numbers (percentages) or medians (interquartile ranges).
Abbreviation: MELD, Model for End-Stage Liver Disease.
*Clinical characteristics between the two groups were compared using the chi-square test for categorical variables or the Mann–Whitney *U*-test for continuous variables.

22.7 kg/m ^{2} ，Child-Pugh评分为7分，MELD评分为9分。总计127例（53%）和69例（29%）患者分别存在多药联用和肌少症。每位患者使用的药物中位数为6种（IQR，4-8，表1）。药物数量随年龄显著增加（P < 0.037；图 1a）。同样，多药联用的患病率也随年龄增长而增加（P = 0.153；图1b）。所用药物类型见表S1。7例接受β受体阻滞剂预防静脉曲张出血的患者被归类为抗高血压组。

总体而言，148例（62%）和91例（28%）患者分别被归类为老年组和年轻成人组（表 1）。与年轻成人组相比，老年组患者中多药联用、高血清肌酐水平及病毒性肝硬化患者比例更高，但酒精性肝硬化患者较少。老年组患者的身高和体重显著低于年轻成人组，但体重指数（BMI）组间无显著差异。老年组的握力和标准肌肉指数（SMI）显著低于年轻成人组，且老年组中肌肉减少症的患病率更高（37% vs 15%，P < 0.001；表 1））。肌少症患者病率随年龄增长而增加（P < 0.001；图 1c）。

多药联用与肌少症在肝硬化患者中的应用

肌少症患病率在多药联用患者中显著高于非多药联用患者（36% vs 21%；P=0.010）。同样，肌少症患者的多药联用率也明显高于非肌少症患者（67% vs 48%；P=0.010）。两组患者在年龄较大、用药量增加、肝功能不全、握力及上肢肌力（SMI）较低等特征上具有相似性（表 2）。

多药联用与肉瘤之间的关联

肌少症。肌少症患者多药治疗的未调整比值比为2.20（95% CI， 1.23-3.94；P=0.008）（表 3）。在调整年龄、性别、体重指数、肝硬化病因和MELD评分后，多药治疗仍与肌少症显著相关（OR，2.11；95%CI， 1.07-4.17；P=0.032）。该关联在老年群体中更为明显（OR，2.62；

表1 肝硬化患者基线特征

特征的	总队列 ($n = 239$)	年轻成人 ($n = 91$)	老年人 ($n = 148$)	P值 *
年龄，岁	68 (58–76)	55 (50–60)	74 (70–79)	<0.001
男人	125 (52)	53 (58)	72 (49)	0.182
用药数量	6 (4–8)	5 (4–7)	7 (3–9)	0.069
多药房	127 (53)	41 (45)	86 (58)	0.049
合并症				
糖尿病	66 (28)	23 (25)	43 (29)	0.555
高血压	151 (63)	51 (56)	100 (68)	0.097
血脂异常	6 (3)	2 (2)	4 (3)	0.809
肝硬化病因				
乙型肝炎病毒	17 (7)	3 (3)	14 (10)	0.002
丙型肝炎病毒	68 (29)	19 (21)	49 (33)	
酒精中毒	60 (25)	34 (37)	26 (18)	
其他	94 (39)	35 (39)	59 (40)	
腹水	113 (47)	44 (48)	69 (47)	0.894
肝性脑病	23 (10)	7 (8)	16 (11)	0.503
MELD评分	9 (7–12)	10 (8–12)	9 (7–12)	0.155
Child-Pugh评分	7 (5–9)	7 (6–9)	7 (5–9)	0.525
白蛋白，g/dl	3.2 (2.6–3.7)	3.2 (2.5–3.5)	3.2 (2.6–3.8)	0.454
肌酐，mg/dl	0.71 (0.60–0.90)	0.65 (0.52–0.78)	0.74 (0.63–0.93)	0.002
钠，mEq/L	138 (136–140)	138 (137–140)	138 (136–140)	0.401
总胆红素，mg/dl	1.2 (0.8–1.7)	1.3 (0.9–1.9)	1.1 (0.8–1.6)	0.057
国际标准化比值	1.12 (1.03–1.26)	1.18 (1.04–1.31)	1.10 (1.02–1.23)	0.008
体成分				
身高，cm	158 (150–166)	164 (158–171)	155 (149–161)	<0.001
体重，kg	58.3 (50.3–65.9)	63.7 (53.9–74.6)	56.0 (48.8–62.5)	<0.001
体重指数，kg/m ^{2}	22.7 (21.4–25.4)	22.73 (21.8–26.3)	22.7 (21.0–25.1)	0.238
握力，kg	20.9 (16.5–29.2)	26.8 (19.7–35.5)	18.6 (14.5–24.9)	<0.001
骨骼肌指数，cm ^{2} /m ^{2}	41.9 (36.2–49.1)	44.9 (38.1–50.0)	40.1 (35.7–47.8)	0.011
肌肉减少症	69 (29)	14 (15)	55 (37)	<0.001

注：数值以数字（百分比）或中位数（四分位距）表示。
缩写：MELD，终末期肝病模型。

采用卡方检验对分类变量进行两组间临床特征比较，连续变量则采用Mann-Whitney U检验。

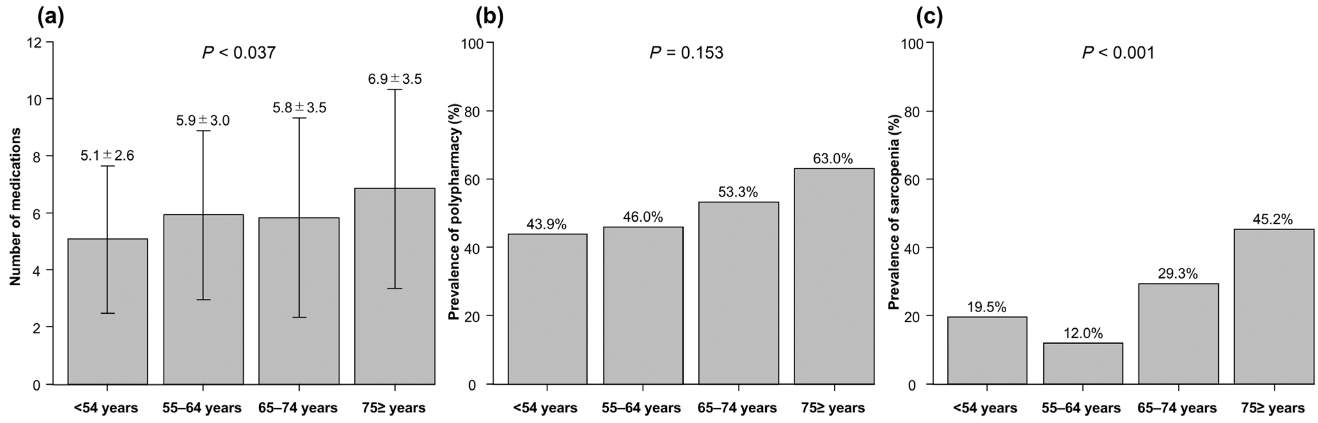

Figure 1 (a) Number of medications used, (b) prevalence of polypharmacy, and (c) prevalence of sarcopenia in 239 patients with cirrhosis in different age groups. Data are compared using the chi-square test and Kruskal–Wallis test.

Table 2 Clinical characteristics of patients with cirrhosis, sarcopenia, and polypharmacy

Characteristic	No polypharmacy ($n = 112$)	Polypharmacy ($n = 127$)	P-value*	No sarcopenia ($n = 170$)	Sarcopenia ($n = 69$)	P-value*
Age, years	67 (57–74)	71 (60–77)	0.010	66 (57–74)	74 (68–78)	<0.001
Men	66 (59)	59 (47)	0.069	88 (52)	37 (54)	0.886
Number of medications used	3 (2–4)	8 (7–10)	<0.001	5 (3–8)	8 (5–10)	<0.001
Polypharmacy	0 (0)	127 (100)	<0.001	81 (48)	46 (67)	0.010
Etiology of cirrhosis						
Hepatitis B virus	7 (6)	10 (8)	0.788	13 (8)	4 (6)	0.750
Hepatitis C virus	29 (26)	39 (31)		50 (29)	18 (26)	
Alcoholic	30 (27)	30 (24)		44 (26)	16 (23)	
Others	46 (41)	48 (38)		63 (37)	31 (45)	
Ascites	26 (23)	87 (69)	<0.001	71 (42)	42 (61)	0.010
Hepatic encephalopathy	2 (2)	21 (17)	<0.001	13 (8)	10 (15)	0.144
MELD score	8 (7–10)	10 (8–13)	<0.001	9 (7–12)	9 (7–12)	0.897
Child–Pugh score	6 (5–8)	8 (7–10)	<0.001	7 (5–9)	8 (6–10)	0.001
Albumin, g/dl	3.5 (3.1–3.9)	2.9 (2.3–3.3)	<0.001	3.3 (2.9–3.7)	2.6 (2.2–3.6)	0.001
Creatinine, mg/dl	0.70 (0.58–0.83)	0.73 (0.60–0.96)	0.083	0.70 (0.59–0.88)	0.75 (0.63–0.92)	0.077
Sodium, mEq/L	139 (138–140)	137 (135–139)	<0.001	138 (137–140)	138 (135–140)	0.022
Total bilirubin, mg/dl	1.1 (0.8–1.6)	1.3 (0.9–1.7)	0.106	1.2 (0.9–1.7)	1.1 (0.8–1.6)	0.571
International normalized ratio	1.07 (1.02–1.19)	1.16 (1.05–1.31)	<0.001	1.12 (1.04–1.25)	1.15 (1.01–1.29)	0.877
Body composition						
Height, cm	160 (152–166)	158 (150–165)	0.150	160 (152–166)	155 (149–162)	0.039
Weight, kg	60.0 (51.6–66.4)	56.6 (49.4–65.3)	0.130	61.2 (52.3–68.4)	53.5 (47.7–58.3)	<0.001
Body mass index, kg/m^2	22.9 (21.0–25.5)	22.4 (21.4–25.1)	0.713	23.6 (22.0–25.9)	21.7 (19.4–23.1)	<0.001
Grip strength, kg	25.1 (18.6–34.1)	18.5 (14.8–24.9)	<0.001	24.5 (18.5–32.5)	16.5 (14.0–19.8)	<0.001
Skeletal muscle index, cm^2/m^2	43.6 (37.9–49.7)	40.5 (35.4–48.0)	0.012	46.0 (41.4–51.3)	34.9 (32.0–37.1)	<0.001
Sarcopenia	23 (21)	46 (36)	0.010	0 (0)	69 (100)	<0.001

Note: Values are presented as numbers (percentages) or medians (interquartile ranges).
Abbreviation: MELD, Model for End-Stage Liver Disease.
*Clinical characteristics between the two groups were compared using the chi-square test for categorical variables or the Mann–Whitney *U*-test for continuous variables.

95% CI, 1.14–6.02; $P = 0.023$) than in the younger adult group (OR, 1.56; 95% CI, 0.40–6.01; $P = 0.522$).

Overall survival. During a median follow-up of 2.2 years (IQR, 0.9–4.0), 62 (26%) patients died. None of the patients underwent liver transplantation during follow-up. OS was not significantly different between the groups ($P = 0.324$, Fig. 2a), with an HR of 1.30 (95% CI, 0.77–2.20; $P = 0.325$). The causes of death are listed in Table S2.

OS was lower in patients with sarcopenia than those without sarcopenia ($P = 0.002$; Fig. 2(b)), with an HR of 2.20 (95% CI, 1.32–3.67; $P = 0.003$). Similarly, OS was lower in patients

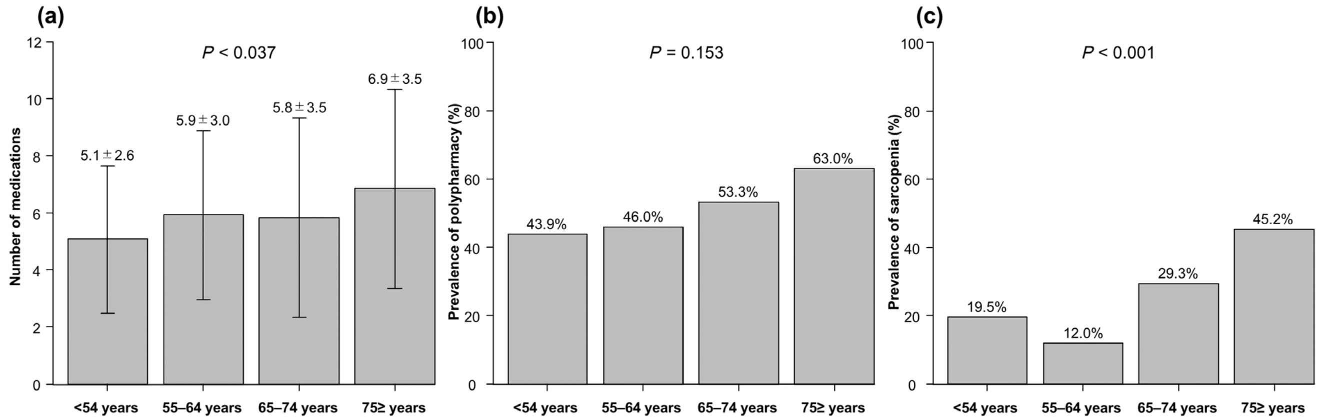

图1 (a) 不同年龄组239例肝硬化患者用药数量、(b) 多药联用率及 (c) 肌少症患病率。数据采用卡方检验和Kruskal-Wallis检验进行比较。

表2 肝硬化、肌少症及多重用药患者的临床特征

特征的	无多重用药 (n = 112)	多药房 (n = 127)	P值 *	无肌少症 (n = 170)	肌肉减少症 (n = 69)	P值 *
年龄，岁	67 (57–74)	71 (60–77)	0.010	66 (57–74)	74 (68–78)	<0.001
男人	66 (59)	59 (47)	0.069	88 (52)	37 (54)	0.886
用药数量	3 (2–4)	8 (7–10)	<0.001	5 (3–8)	8 (5–10)	<0.001
多药房	0 (0)	127 (100)	<0.001	81 (48)	46 (67)	0.010
肝硬化病因						
乙型肝炎病毒	7 (6)	10 (8)	0.788	13 (8)	4 (6)	0.750
丙型肝炎病毒	29 (26)	39 (31)		50 (29)	18 (26)	
酒精中毒	30 (27)	30 (24)		44 (26)	16 (23)	
其他	46 (41)	48 (38)		63 (37)	31 (45)	
腹水	26 (23)	87 (69)	<0.001	71 (42)	42 (61)	0.010
肝性脑病	2 (2)	21 (17)	<0.001	13 (8)	10 (15)	0.144
MELD评分	8 (7–10)	10 (8–13)	<0.001	9 (7–12)	9 (7–12)	0.897
Child-Pugh评分	6 (5–8)	8 (7–10)	<0.001	7 (5–9)	8 (6–10)	0.001
白蛋白，g/dl	3.5 (3.1–3.9)	2.9 (2.3–3.3)	<0.001	3.3 (2.9–3.7)	2.6 (2.2–3.6)	0.001
肌酐，mg/dl	0.70 (0.58–0.83)	0.73 (0.60–0.96)	0.083	0.70 (0.59–0.88)	0.75 (0.63–0.92)	0.077
钠，mEq/L	139 (138–140)	137 (135–139)	<0.001	138 (137–140)	138 (135–140)	0.022
总胆红素，mg/dl	1.1 (0.8–1.6)	1.3 (0.9–1.7)	0.106	1.2 (0.9–1.7)	1.1 (0.8–1.6)	0.571
国际标准化比值	1.07 (1.02–1.19)	1.16 (1.05–1.31)	<0.001	1.12 (1.04–1.25)	1.15 (1.01–1.29)	0.877
体成分						
身高，cm	160 (152–166)	158 (150–165)	0.150	160 (152–166)	155 (149–162)	0.039
体重，kg	60.0 (51.6–66.4)	56.6 (49.4–65.3)	0.130	61.2 (52.3–68.4)	53.5 (47.7–58.3)	<0.001
体重指数，kg/m ^{2}	22.9 (21.0–25.5)	22.4 (21.4–25.1)	0.713	23.6 (22.0–25.9)	21.7 (19.4–23.1)	<0.001
握力，kg	25.1 (18.6–34.1)	18.5 (14.8–24.9)	<0.001	24.5 (18.5–32.5)	16.5 (14.0–19.8)	<0.001
骨骼肌指数，cm ^{2} /m ^{2}	43.6 (37.9–49.7)	40.5 (35.4–48.0)	0.012	46.0 (41.4–51.3)	34.9 (32.0–37.1)	<0.001
肌肉减少症	23 (21)	46 (36)	0.010	0 (0)	69 (100)	<0.001

注：数值以数字（百分比）或中位数（四分位距）表示。
缩写：MELD，终末期肝病模型。
采用卡方检验对分类变量进行两组间临床特征比较，连续变量则采用Mann-Whitney U检验。

与年轻成人组相比，该组的OR为1.56（95% CI， 0.40-6.01；P = 0.522）（95% CI， 1.14-6.02；P = 0.023）。

总生存期。中位随访时间为2.2年（IQR，0.9-4.0），62例（26%）患者死亡。随访期间无患者接受肝移植。总生存期（OS）未见变化。组间差异无统计学意义（P=0.324，图 2a），风险比（HR）为1.30（95%CI， 0.77-2.20；P=0.325）。死亡原因列于表S2。

肌少症患者的OS低于无肌少症患者（P=0.002；图 2(b)），HR为2.20（95%CI， 1.32-3.67；P=0.003）。同样，

Table 3 Association between sarcopenia and polypharmacy

	Total cohort OR (95% CI)	*P*-value	Younger adults OR (95% CI)	*P*-value	Older adults OR (95% CI)	*P*-value
Polypharmacy unadjusted	2.20 (1.23–3.94)	0.008	1.78 (0.56–5.62)	0.327	2.10 (1.04–4.23)	0.039
Polypharmacy adjusted for age	1.93 (1.06–3.51)	0.032	1.79 (0.56–5.66)	0.916	1.97 (0.97–4.01)	0.061
Polypharmacy adjusted for age and sex	2.01 (1.10–3.70)	0.024	1.91 (0.59–6.12)	0.279	2.03 (0.99–4.15)	0.053
Polypharmacy adjusted for age, sex, and BMI	2.02 (1.06–3.84)	0.032	2.17 (0.63–7.50)	0.221	1.95 (0.91–4.15)	0.085
Polypharmacy adjusted for age, sex, BMI, and etiology	2.10 (1.09–4.04)	0.026	2.12 (0.60–7.56)	0.245	2.23 (1.01–4.90)	0.046
Polypharmacy adjusted for age, sex, BMI, etiology, and MELD score	2.11 (1.07–4.17)	0.032	1.56 (0.40–6.01)	0.522	2.62 (1.14–6.02)	0.023

Abbreviations: BMI, body mass index; CI, confidence interval; MELD, Model for End-Stage Liver Disease; OR, odds ratio.

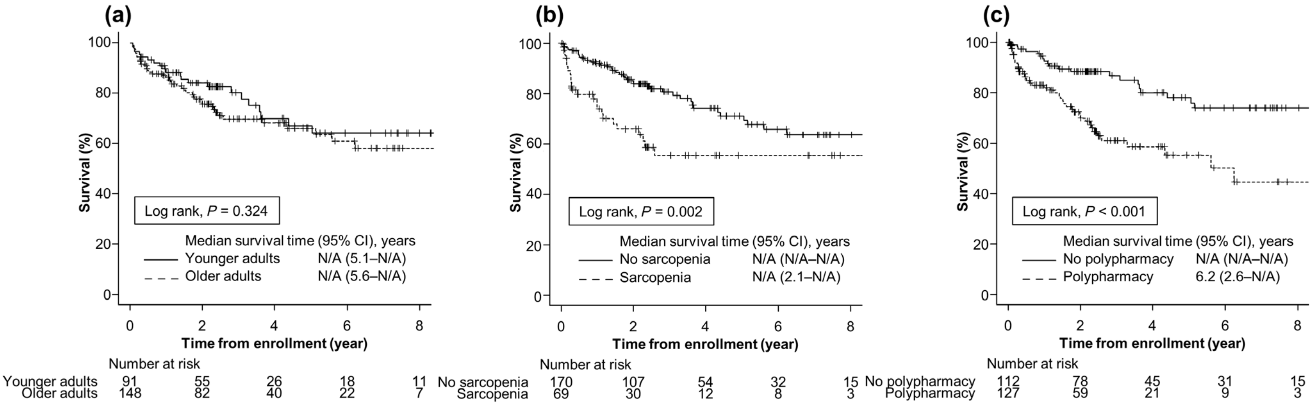

Figure 2 Overall survival in (a) older and younger adults, (b) patients with and without sarcopenia, and (c) patients with and without polypharmacy. Overall survival is estimated using the Kaplan–Meier method and compared between groups using the log-rank test.

Table 4 Prognostic value of sarcopenia and polypharmacy in multivariate analyses*

	Total cohort HR (95% CI)	*P*-value	Younger adults HR (95% CI)	*P*-value	Older adults HR (95% CI)	*P*-value
Sarcopenia	2.00 (1.12–3.50)	0.020	2.57 (0.82–7.44)	0.102	1.71 (0.83–3.51)	0.148
Polypharmacy	1.83 (1.01–3.37)	0.045	1.47 (0.54–4.01)	0.444	2.31 (1.01–5.67)	0.046
MELD score	1.28 (1.19–1.38)	<0.001	1.30 (1.15–1.47)	<0.001	1.26 (1.14–1.40)	<0.001

Abbreviations: CI, confidence interval; HR, hazard ratio; MELD, Model for End-Stage Liver Disease.
*Adjusted for age, sex, body mass index, and etiology.

with polypharmacy than in those without polypharmacy ($P < 0.001$; Fig. 2c), with an HR of 2.75 (95% CI, 1.60–4.71; $P < 0.001$).

In subgroup analysis, sarcopenia was significantly associated with poor prognosis in the younger adult group ($P = 0.026$; Fig. S1a), with an HR of 2.81 (95% CI, 1.09–7.28; $P = 0.033$), and in the older adult group ($P = 0.045$; Fig. S1b), with an HR of 1.87 (95% CI, 1.01–3.49; $P = 0.048$). However, polypharmacy was significantly associated with poor prognosis in the older adult group ($P < 0.001$; Fig. S2b), with an HR of 3.80 (95% CI, 1.79–8.05; $P < 0.001$), but not in the younger adult group ($P = 0.265$; Fig. S2a), with an HR of 1.63 (95% CI, 0.68–3.90; $P = 0.269$).

Prognostic significance of polypharmacy and sarcopenia in patients with cirrhosis. The prognostic factors of patients with cirrhosis evaluated by multivariate analysis are listed in Table 4. After adjusting for confounders, sarcopenia (HR, 2.00; 95% CI, 1.12–3.50; $P = 0.020$), polypharmacy (HR, 1.83; 95% CI, 1.01–3.37; $P = 0.045$), and MELD score (HR, 1.28; 95% CI, 1.19–1.38; $P < 0.001$) were significantly associated with mortality.

In younger adults, MELD score was significantly associated with mortality (HR, 1.30; 95% CI, 1.15–1.47; $P < 0.001$). In older adults, polypharmacy (HR, 2.31; 95% CI, 1.01–5.67; $P = 0.046$) and MELD score (HR, 1.26; 95% CI, 1.14–1.40; $P < 0.001$), but not sarcopenia (HR, 1.71; 95% CI, 0.83–3.51;

表3 肌少症与多重用药之间的关联

	总队列 或（95% CI）	概率值	年轻成人 或（95% CI）	概率值	老年人 或（95% CI）	概率值
未调整的多重用药	2.20 (1.23–3.94)	0.008	1.78 (0.56–5.62)	0.327	2.10 (1.04–4.23)	0.039
经年龄校正的多重用药	1.93 (1.06–3.51)	0.032	1.79 (0.56–5.66)	0.916	1.97 (0.97–4.01)	0.061
经年龄和性别校正的多重用药	2.01 (1.10–3.70)	0.024	1.91 (0.59–6.12)	0.279	2.03 (0.99–4.15)	0.053
经年龄、性别和体重指数（BMI）校正的多重用药	2.02 (1.06–3.84)	0.032	2.17 (0.63–7.50)	0.221	1.95 (0.91–4.15)	0.085
经年龄、性别、体重指数（BMI）及病因校正的多重用药	2.10 (1.09–4.04)	0.026	2.12 (0.60–7.56)	0.245	2.23 (1.01–4.90)	0.046
经年龄、性别、体重指数（BMI）、病因学及MELD评分校正的多重用药情况	2.11 (1.07–4.17)	0.032	1.56 (0.40–6.01)	0.522	2.62 (1.14–6.02)	0.023

缩写：BMI，体重指数；CI，置信区间；MELD，终末期肝病模型；OR，比值比。

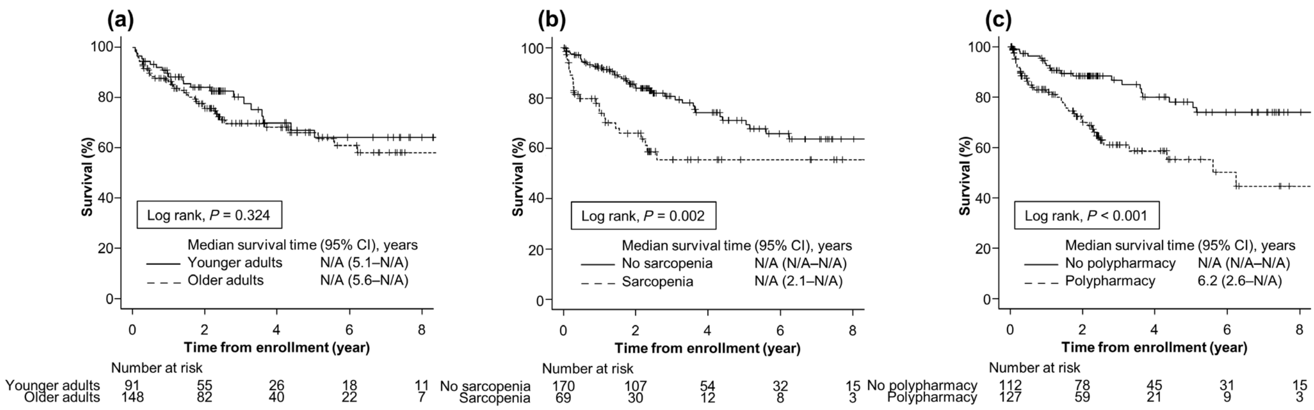

图2 总体生存率在(a)老年与年轻成人、(b)肌少症患者与非肌少症患者、(c)多药治疗患者与非多药治疗患者中的比较。采用Kaplan-Meier法估算总体生存率，并通过对数秩检验进行组间比较。

表4多变量分析中肌少症和多重用药的预后价值 *

	总队列 HR（95% CI）	概率值	年轻成人 HR（95% CI）	概率值	老年人 HR（95% CI）	概率值
肌肉减少症	2.00 (1.12–3.50)	0.020	2.57 (0.82–7.44)	0.102	1.71 (0.83–3.51)	0.148
多药房	1.83 (1.01–3.37)	0.045	1.47 (0.54–4.01)	0.444	2.31 (1.01–5.67)	0.046
MELD评分	1.28 (1.19–1.38)	<0.001	1.30 (1.15–1.47)	<0.001	1.26 (1.14–1.40)	<0.001

缩写：CI，置信区间；HR，风险比；MELD，终末期肝病模型。
*已根据年龄、性别、体重指数及病因进行校正。

与未合并用药者相比，合并用药者发生率显著增高（P < 0.001；图 2 c），风险比（HR）为2.75（95%CI， 1.60-4.71 ；P < 0.001）。

在亚组分析中，肌少症与年轻成人组的不良预后显著相关（P = 0.026；图 S1a），风险比为 2.81（95% CI， 1.09-7.28 ；P = 0.033），与老年成人组的不良预后也显著相关（P = 0.045；图 S1b），风险比为 1.87（95% CI， 1.01-3.49 ；P = 0.048）。然而，多药治疗与老年成人组的不良预后显著相关（P < 0.001；图 S2b），风险比为 3.80（95% CI， 1.79-8.05 ；P < 0.001），但在年轻成人组中未见显著相关（P = 0.265；图 S2a），风险比为 1.63（95% CI， 0.68-3.90 ；P = 0.269）。

多药联用与SAR的预后意义

肝硬化患者中肌肉减少症。通过多变量分析评估的肝硬化患者的预后因素列于表 4 4。在调整混杂因素后，肌肉减少症（HR，2.00；95%CI， 1.12-3.50 ；P=0.020）、多药治疗（HR，1.83；95%CI， 1.01-3.37 ；P=0.045）和MELD评分（HR，1.28；95%CI， 1.19-1.38 ；P<0.001）与死亡率显著相关。

在年轻成人中，MELD 评分与死亡率显著相关（HR，1.30；95% CI， 1.15-1.47 ；P < 0.001）。在老年成人中，多药治疗（HR，2.31；95% CI， 1.01-5.67 ；P = 0.046）和 MELD 评分（HR，1.26；95% CI， 1.14-1.40 ；P < 0.001）与死亡率相关，但肌肉减少症（HR，1.71；95% CI， 0.83-3.51 ；

$P = 0.148$), were significantly associated with mortality. In addition, the impact of polypharmacy and sarcopenia on mortality was compared in two groups, young and old. The results showed that polypharmacy (HR, 2.53; 95% CI, 1.48–4.47) and sarcopenia (HR, 1.92; 95% CI, 1.11–3.25) were independently associated mortality.

Discussion

This study, to our knowledge, is the first to evaluate the significance of polypharmacy and sarcopenia in patients with cirrhosis. First, the prevalence of polypharmacy and sarcopenia was high in these patients and increased with age. Second, patients with polypharmacy and sarcopenia had similar characteristics, such as older age, increased medication use, advanced liver disease, and decreased skeletal muscle mass and strength. Third, there was a strong association between polypharmacy and sarcopenia, especially in the older adult group. Fourth, these two factors adversely affected prognosis; more importantly, the prognostic significance of polypharmacy was more pronounced in older adults than in younger adults. It is difficult to distinguish between appropriate polypharmacy, which has health benefits, and inappropriate polypharmacy, in which multiple medications are no longer required.[2] However, our results suggest that polypharmacy and sarcopenia are important predictors of prognosis and that mitigating the impact of polypharmacy on sarcopenia through the appropriate use of medications improves survival, especially in older adults with cirrhosis.

Polypharmacy increases the likelihood of adverse events, including drug–drug interactions, poor medication adherence, falls, cognitive impairment, and death.[1,15] A nationwide survey in Japan showed that the prevalence of polypharmacy was 28% and 41% in the population aged 65–74 and ≥75 years, respectively.[15] In our cohort, the prevalence of polypharmacy was higher in these two groups, corresponding to 53.3% and 63%, respectively, demonstrating that polypharmacy is common in patients with cirrhosis. Moreover, 53% of patients with cirrhosis had polypharmacy, consistent with the rate reported in a cross-sectional study of patients with cirrhosis (56%).[10] Aging commonly involves morphological and functional changes in the liver, including decreased liver volume, reduced blood flow, and impaired metabolic and detoxification functions.[1,8,9] Moreover, patients with cirrhosis are at an increased risk of altered drug pharmacokinetics because of pathophysiological changes such as decreased metabolic capacity, lower albumin synthesis, and portosystemic shunt.[6,10,13] Recent evidence shows that 28% and 22% patients with cirrhosis have adverse drug reactions and drug–drug interactions, respectively.[13] Thus, these patients are more prone to adverse reactions caused by polypharmacy than those without cirrhosis, and more attention should be paid to optimizing the number of medications to reduce the risk of adverse events in patients with cirrhosis.[10]

Polypharmacy and sarcopenia are common geriatric syndromes; however, their interaction is complex.[1] The present study found that patients with polypharmacy had a twofold higher OR for sarcopenia, supporting the findings of previous studies.[1,17] Several mechanisms may underlie the association between polypharmacy and sarcopenia. Medications that cause side effects, including dysphagia and gastrointestinal symptoms, can lead to poor nutritional status and sarcopenia.[1,18] In addition, medications for liver disease, such as loop diuretics, proton pump inhibitors, and glucocorticoids, can have detrimental effects on skeletal muscle.[1,19–23] In the present study, patients with polypharmacy used loop diuretics, proton pump inhibitors, and glucocorticoids more frequently than those without polypharmacy. In addition, several medications (e.g., statins and targeted anticancer agents) are known to contribute to the pathogenesis of sarcopenia through mitochondrial dysfunction, activation of the transcription factor NF-kappa B, and increases in reactive oxygen species and inflammatory cytokines.[1] The insights gained from this study may help determine the extent to which these medications contribute to sarcopenia in cirrhosis.

Polypharmacy and sarcopenia worsen prognosis in older adults[1]; however, there is limited evidence on the prognostic significance of these factors in patients with cirrhosis. In our study, polypharmacy and sarcopenia independently predicted mortality. Furthermore, the prognostic significance of polypharmacy was stronger in older adults than in younger adults. This result might be attributed to the higher incidence of adverse drug reactions, geriatric syndromes, functional and cognitive impairments, and multimorbidity in the former group.[13,24] Therefore, assessment of polypharmacy may be useful to stratify the risk of mortality in patients with cirrhosis.[9] However, the ideal timing and frequency of polypharmacy assessment and effect of polypharmacy on the health outcomes of these patients need to be investigated.[25–27]

This study has two positive points. First, the number of participants ($N = 239$) and long follow-up period (minimum 2 years) may allow us to estimate adjusted ORs for sarcopenia and HRs for death. It is well known that as the age increases, the number of comorbidities and therefore treatment (drugs) increases. The findings of our study need validation in large population-based prospective studies. Second, the analyses were adjusted for important confounders in the association between polypharmacy and sarcopenia or mortality.

This study has limitations. First, the observational design does not allow for the identification of adverse drug reactions, drug–drug interactions, or drug–disease interactions, which may affect the interpretation of the results. In addition, the findings may be influenced by residual confounding and clinically relevant factors such as daily physical activity, comorbidities, cognitive impairment, and malnutrition.[17,18] Second, polypharmacy and sarcopenia were assessed at a single time point, preventing the evaluation of changes over time. Third, data on nonprescribed medications, such as dietary supplements, over-the-counter medications, and traditional and complementary medicines that may cause adverse events or interactions with prescribed medications,[2] were not collected and analyzed. Fourth, the definition of polypharmacy and diagnostic criteria for sarcopenia vary widely.[1,28] Considering that the diagnoses of polypharmacy and sarcopenia were based on Japanese guidelines,[14–16] our results might not be extrapolated to other cohorts in different regions and clinical settings. Thus, these findings should be interpreted with caution and validated in a large external cohort. However, we believe that this study serves as a basis for future research.

Conclusion

In conclusion, our results provide convincing evidence that polypharmacy and sarcopenia are common, interconnected, and useful in predicting mortality in patients with cirrhosis. This study underscores the importance of avoiding inappropriate polypharmacy, P = 0.148），与死亡率显著相关。此外，比较了多药联用和肌肉减少症对年轻组和老年组死亡率的影响。结果显示，多药联用（HR，2.53；95% CI， 1.48-4.47）和肌肉减少症（HR，1.92；95% CI， 1.11-3.25）与死亡率独立相关。

讨论

据我们所知，本研究首次评估了肝硬化患者中多药联用与肌肉减少症的临床意义。首先，这些患者中多药联用和肌肉减少症的患病率较高，且随年龄增长而增加。其次，多药联用和肌肉减少症患者具有相似特征，如高龄、用药量增加、肝病进展、骨骼肌质量与力量下降。第三，多药联用与肌肉减少症存在显著关联，尤其在老年患者群体中更为明显。第四，这两个因素对预后产生不利影响；更重要的是，多药联用对老年患者的预后影响显著大于年轻患者。目前难以区分具有健康益处的合理多药联用与不再需要的不合理多药联用。[2] 然而，我们的研究结果表明，多药联用和肌肉减少症是预后的关键预测因素，通过合理用药减轻多药联用对肌肉减少症的影响，可提高生存率，尤其对老年肝硬化患者具有重要意义。

多重用药会显著增加不良事件发生风险，包括药物相互作用、用药依从性差、跌倒、认知功能障碍乃至死亡。日本[1,15] 一项全国调查显示，65-74岁和75岁及以上人群的多重用药率分别为28%和41%。[15] 在我们的研究队列中，这两个年龄段的多重用药率更高，分别为53.3%和63%，这表明肝硬化患者普遍存在多重用药现象。值得注意的是，53%的肝硬化患者存在多重用药，与一项横断面研究报道的56%发生率基本吻合。随着[10] 衰老进程，肝脏通常会出现形态和功能改变，具体表现为肝体积缩小、血流减少以及代谢和解毒功能下降。[1,8,9] 此外，由于代谢能力下降、白蛋白合成减少及门体分流等病理生理变化，肝硬化患者发生药物代谢动力学改变的风险也会显著增加。[6,10,13] 最新数据显示，28%的肝硬化患者会出现药物不良反应，22%会出现药物相互作用。[13] 由此可见，这类患者比非肝硬化人群更容易因多重用药引发不良反应，因此需要特别注意优化用药方案，减少药物数量，从而降低肝硬化患者的不良事件风险。[10]

多药联用与肌肉减少症是老年群体中常见的综合征，但二者间的相互作用机制较为复杂。[1] 本研究发现，多药联用患者的肌肉减少症发生几率是普通患者的两倍，这一结果与既往研究结论相吻合。[1,17] 多种机制可能是导致多药联用与肌肉减少症关联的原因：引发吞咽困难、胃肠道不适等副作用的药物，可能造成营养状况恶化并诱发肌肉减少症；[1,18] 此外，治疗肝病的药物（如祥利尿剂、质子泵抑制剂和糖皮质激素）也可能产生影响。多药联用可能对骨骼肌产生不良影响。[1,19–23] 本研究发现，与未联用药物的患者相比，多药联用患者更频繁地使用袢利尿剂、质子泵抑制剂和糖皮质激素。此外，他汀类药物和靶向抗癌药物等几种药物，已知会通过线粒体功能障碍、激活转录因子 NF -κB，以及增加活性氧和炎症细胞因子来促进肌少症的发生。[1] 本研究获得的见解，可能有助于评估这些药物在肝硬化患者肌少症发病机制中的具体作用程度。

多药共用和肌肉减少症会加重老年患者的预后[1]，但关于这些因素对肝硬化患者预后影响的证据仍显不足。本研究发现，多药共用和肌肉减少症均可独立预测死亡风险。值得注意的是，老年患者使用多种药物的预后风险显著高于年轻群体。这可能与老年患者更易出现药物不良反应、老年综合征、功能与认知障碍以及多重疾病等状况有关。[13,24] 因此，评估多药共用情况或可作为肝硬化患者死亡风险分层的重要指标。[9] 不过，关于多药共用的最佳评估时机、评估频率，以及其对患者健康结局的具体影响，仍有待进一步研究验证。[25–27]

本研究具有两个积极意义。首先，239例受试者样本量与长达2年的随访期，使我们能够估算出肌少症的校正比值比（OR）及死亡风险比（HR）。众所周知，随着年龄增长，合并症数量及相应药物治疗方案会增加。本研究结果需在大规模人群前瞻性研究中进行验证。其次，分析过程中已对多重用药与肌少症或死亡率关联中的重要混杂因素进行了校正。

本研究存在若干局限性。首先，观察性研究设计无法识别药物不良反应、药物间相互作用或药物与疾病间的相互作用，这可能影响研究结果的解读。此外，研究结果可能受到残余混杂因素及临床相关因素（如日常体力活动、合并症、认知障碍和营养不良）的影响。[17,18] 其次，多药共用和肌肉减少症的评估仅在单一时间点进行，导致无法追踪随时间的变化趋势。第三，研究[2] 未收集和分析非处方药物（如膳食补充剂、非处方药及传统与补充医学）的相关数据，这些药物可能引发不良事件或与处方药产生相互作用。第四，多药共用的定义及肌肉减少症的诊断标准存在显著差异。[1,28] 鉴于本研究采用日本指南对多药共用和肌肉减少症进行诊断，[14–16] 其研究结果可能无法推广至其他地区和临床环境中的不同人群。因此，这些发现应谨慎解读，并需在大型外部队列中进行验证。然而，我们认为本研究可为未来研究奠定基础。

结论

综上所述，我们的研究结果提供了令人信服的证据，表明多药联用与肌肉减少症在肝硬化患者中具有普遍性、相互关联性，并可有效预测死亡率。本研究强调了避免不当多药联用的重要性。

especially for patients with cirrhosis, and the need to implement other strategies to ensure appropriate pharmacotherapy and prevent or delay sarcopenia and related adverse events.

Informed consent statement. Informed consent was obtained from all subjects involved in the study.

Data availability statement. The data presented in this study are available upon request from the corresponding author.

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Supporting information

Additional supporting information may be found in the online version of this article at the publisher's website:

Figure S1. Overall survival of patients with and without sarcopenia in the (a) younger adult group and (b) older adult group. Overall survival is estimated using the Kaplan–Meier method and compared between groups using the log-rank test.

Figure S2. Overall survival of patients with and without polypharmacy in the (a) younger adult group and (b) older adult group. Overall survival is estimated using the Kaplan–Meier method and compared between groups using the log-rank test.

Table S1. Types of medications used by patients†
Table S2. Causes of death in our cohort

尤其对于肝硬化患者，需实施其他策略以确保适当药物治疗，并预防或延缓肌少症及相关不良事件的发生。

知情同意声明。知情同意为来自本研究所有受试者。

数据可用性声明。所呈现的数据本研究资料可应要求向通讯作者索取。

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支持信息

更多补充信息可于出版商网站的本文在线版本中查阅：

图S1. (a) 年轻成人组与 (b) 老年成人组中伴或不伴肌少症患者的总生存期。总生存期采用Kaplan-Meier法进行估算，并通过对数秩检验进行组间比较。

图S2. 多药联用与非多药联用患者在(a)年轻成人组和(b)老年成人组中的总生存期。总生存期采用Kaplan-Meier法进行估算，并通过对数秩检验进行组间比较。

表S1.患者使用的药物类型 †
表S2. 我们队列中的死亡原因